

Honeynets: How good are they

Advantages

- Flexibility; any system or application can be placed in a honeynet environment
- Extensive data capture capabilities for both known and unknown tools and tactics
- Adaptable to many organisations and environments

Disadvantages

- Complexity of deployment and resources required to maintain
- High-interaction functionality introduces the risk of attackers using the systems to attack, harm or infiltrate other systems or organisations
- New and immature technologies have a greater risk of breaking or introducing errors

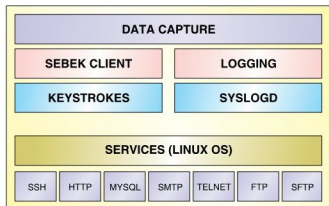


Fig. 3: Honeypot components

Various mechanisms can be implemented to protect against a single point of failure, especially when dealing with new or unknown attacks. Also, data control should operate in a fail-closed manner. This means, if there is a failure in the administrator's mechanisms (process dying, hard drive full, misconfigured rules), the honeynet architecture should block all the outbound activity.

Bridging. As discussed earlier, the honeynet has to be as anonymous as possible to the remote hacker. To achieve this, honeywall is kept hidden by not assigning any IP address to it.

Data capture. It is the monitoring and logging of all of the attacker's

activities within the honeynet. The captured data is then analysed to learn the tools, tactics and motives of attackers. The challenge is to capture as much data as possible without the attacker's knowledge. It is critical to use multiple mecha-

nisms for capturing the data.

Data is captured at various layers. The more the layers of information captured at both the network and host level, the more can be learned. One challenge of data capture is that a large portion of attacker activity happens over encrypted channels (such as IPSec, SSH and SSL).

Data analysis. The entire purpose of a honeynet is gathering information. Honeynets are worthless if they are unable to convert the collected data into information. Security analysts must have some means to analyse the data. Different organisations have different needs, and as such different data analysis requirements.

Data collection. It applies par-

ticularly to organisations that have multiple honeynets in distributed environments. Most organisations will have only one honeynet, called a standalone deployment. Organisations that have multiple honeynets logically or physically distributed around the world have to collect all of the captured data and store it in a central location. This way the captured data can be combined, exponentially increasing its value.

Data collection provides a secure means of centrally storing all of the captured information from distributed honeynets. Honeywall stores most of the fused data (refer Fig. 2) in the database called Hflow—according to The Honeynet Project (<https://projects.honeynet.org/hflow>).

Honeypot

Fig. 3 depicts components of a honeypot. A honeypot is a simple host where an attacker can interact through his malicious activities. To track the activities, data capture, including attacker's logged-in session and keystrokes, is needed at this level too.

Tools for data capture

To track hackers' activities, the analyst must understand honeynet data capture possibilities. Honeynet tools required for data capture at network and host levels are:

Data capture at network level. The hacker's inbound and outbound traffic is gathered while monitoring his activities. The various tools required for network data capture are:

Full network packet data capture (inside/outside). It involves data capture at the bridge layer (or at span/monitored port).

Passive OS fingerprinting (pOf). It in-



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